

Presentation Deck (4 Slides)

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Slide 1: Title & Problem Statement

- **Headline:** KOHLER AI BathPlanner
- **Sub-headline:** Eliminating Design Friction with Deterministic AI
- **Presenter:** Naman Agarwal, Dr. Vishwanath Karad MIT World Peace University
- **The Problem:** Standard Large Language Models hallucinate. They confidently invent non-existent products, ignore physical room clearances, and bypass strict budget limits. This makes generic chatbots dangerous for actual interior design and architectural planning.
- **The Solution:** An interactive design assistant that uses AI solely for natural-language understanding, while delegating all product selection to a deterministic, math-based constraint engine.

Slide 2: System Architecture

- **Headline:** The Hybrid Engine
- **NLP Extraction Layer:** Translates conversational requests into strict JSON parameters (Dimensions, Budget, Aesthetics).
- **Deterministic Constraint Engine (Python):** Exhaustively searches a verified local KOHLER catalog. It mathematically rejects any product combinations that exceed the square footage limit or the financial budget.
- **Weighted Compatibility Scoring:** Feasible combinations are ranked on a 0-100 scale balancing budget fit, space compatibility, style matching, and sustainability.
- **Procedural 3D Visualization:** Generates dynamic WebGL scenes (Three.js) based on the exact output of the constraint engine.

Slide 3: The User Experience

- **Headline:** Seamless Concept to Visualization

- **1. Natural Language Input:** Users describe their space naturally ("8×8 ft, 4 Lakh budget, Japanese Zen...").
- **2. Real-Time Math:** The system calculates the footprint and cost, returning the highest-scoring KOHLER product bundle.
- **3. Interactive 3D Render:** Users instantly explore their selected fixtures in a procedurally generated, style-specific 3D environment featuring dynamic lighting and biophilic decor.
- **4. Conversational Redesign:** Users can type "make it more luxurious," instantly triggering a new calculation and updated 3D render without losing their budget constraints.

Slide 4: Innovation & Business Impact

- **Headline:** Defensible Technology, Measurable Value
- **Brand Protection (Zero Hallucination):** By mathematically isolating the product catalog from the LLM, KOHLER ensures customers are only recommended real, physically viable configurations.
- **Accelerated Product Discovery:** Removes the friction of cross-referencing spec sheets and clearance manuals manually.
- **Built-in Sustainability:** The algorithm actively weights and rewards water-efficient product selections, nudging users toward eco-friendly designs.
- **Premium Visualization:** Upgrades standard text-based recommendations into high-contrast, mathematically scaled 3D environments.